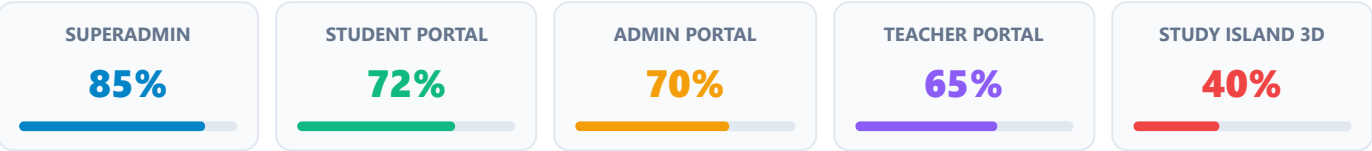


Platform Feature Audit & Completion Roadmap

Exhaustive technical assessment across all 4 portals and 3D Engine modules with phase-by-phase execution plan.

Audited Portals: Student, Teacher, Admin, SuperAdmin, Study Island **Database:** Supabase PostgreSQL + R2 CDN **Architecture:** Zero-Waste Ponytail Discipline

PORTAL HEALTH & COMPLETION MATRIX



DETAILED COMPONENT & FILE AUDIT

COMPONENT / VIEW	PORTAL	SUPABASE CONNECTIVITY	GAPS / PLACEHOLDERS	STATUS
LiveClass.jsx	Teacher	live_classes, notifications, tests	Class dropdown uses mock array	90% Ready
ToDoList.jsx	Teacher	None (0% backend sync)	Local React state only, resets on refresh	25% Stub
SmartboardTeaching.jsx	Teacher	Loads Study Island simulation	Canvas annotation overlay not wired	60% Partial
QuestionBank.jsx	Teacher	question_bank (Add/Delete/Search)	Edit question handler missing	90% Ready
Progress.jsx	Student	Reads localStorage cache	Static trajectory curves, badges locked	35% Stub
Mentors.jsx	Student	teachers, users, Presence WebSocket	Cosmetic 5.0 rating (no reviews table)	90% Ready
Chats.jsx	Student	conversations, messages, announcements	Subtitle has static teacher name	95% Ready
Events.jsx	Admin	announcements, live_classes (Full CRUD)	Attendee counts are hardcoded	95% Ready
Communications.jsx	Admin	announcements, Realtime chat	Class Group messaging tab unwired	90% Ready
Fee Management	Admin	student_fees schema defined	No Fees.jsx view built yet	0% Missing
Study Island 3D Content	3D Engine	Science (Optics & Shadows) Live	5 subjects & 6 chapters show toasts	40% Partial
Mixed Reality Tab	3D Engine	Locked tab state	WebXR 360° device fallback needed	20% Stub

🚀 KEY DEFICIENCIES & CORRECTIVE ACTION PLAN

1. Teacher ToDo List Persistence

Teacher Portal/src/views/ToDoList.jsx

Currently operates exclusively in volatile React component state. All created items and completed states disappear on navigation or browser refresh.

Fix: Wire ToDo items to Supabase todos table scoped by user_id with real-time optimistic updates.

2. Student Academic Progress & Achievement Badges

Student portal/src/views/Progress.jsx

Recharts curve uses static mock data arrays. Achievement badges (Optics Master, Speed Solver, 10-Day Streak) remain permanently locked.

Fix: Connect progress calculations to real quiz submission scores (student_test_results / quiz_scores) and unlock badges when criteria are met.

3. Teacher Smartboard Interactive Drawing Canvas

Teacher Portal/src/views/SmartboardTeaching.jsx

Interactive 3D simulation launches cleanly in the smartboard frame, but overlay drawing tools (Pen, Laser, Highlighter, Eraser) do not render on a live canvas.

Fix: Mount an HTML5 transparent overlay canvas above the 3D frame with smooth stroke drawing and snapshot saving.

4. Admin Portal Fee Management & Receipts

Admin Portal/src/views/Fees.jsx [NEW]

Tuition and laboratory fee tracking is not yet exposed in the Admin Portal navigation despite having backend database schema definitions.

Fix: Build Fees.jsx view with fee collection modal, payment method tracking (Cash/Online), and instant downloadable PDF receipts.

5. Study Island Queued 3D Chapters & Subjects

study-island/script.js (lines 1550-1600)

Mathematics, Geography, History, and 6 Science chapters (Skeletal System, Pollination, Fertilization, Solids) trigger toast placeholders.

Fix: Build 3D spatial geometry interactive engine, Geography 3D Earth globe, and 3D Skeletal anatomy simulation.

🚀 STRATEGIC PHASED IMPLEMENTATION ROADMAP

⚡ Phase 1: High-Impact Quick Wins (Immediate)

1-2 Days • Zero New UI

- 1 **Persist Teacher ToDo List:** Wire ToDoList.jsx to Supabase PostgreSQL table.
- 2 **Dynamic Class Selectors:** Populate class dropdowns in LiveClass.jsx & Inbox.jsx from classes table.
- 3 **Real Badge Unlocking:** Calculate badge unlock triggers dynamically in Progress.jsx based on quiz scores.
- 4 **Dynamic Student Chat Header:** Resolve assigned mentor/teacher name dynamically in Chats.jsx.
- 5 **Live Event Attendee Metrics:** Calculate real attendance figures in Events.jsx.

🔧 Phase 2: Core Functional Modules (Near Term)

3-5 Days • Full DB Integration

- 1 **Smartboard Drawing Canvas:** Build HTML5 transparent drawing overlay with pen & laser tools.
- 2 **Admin Fee Management:** Implement Fees.jsx view with receipt generator and DB records.
- 3 **Admin Forensic Audit Log:** Stream system_audit_logs into Admin security tab.
- 4 **Student Test Result Sync:** Save test submission grades to Supabase backend table.

🌐 Phase 3: 3D Content Expansion & WebXR (Scale)

1-2 Weeks • Three.js 3D WebGL

- 1 **3D Mathematics Spatial Geometry:** Interactive Three.js volume & angle visualizer.
- 2 **3D Anatomy & Skeletal Explorer:** Click-to-inspect bone structure & anatomical functions.
- 3 **Geography 3D Earth Engine:** Interactive continents, climate zones, and tectonic globe.

